A NOETHERIAN REGULAR RING OF FINITE DIMENSION IS A STRONG GENERATOR FOR ITS PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(small detail omitted)' once inside the counted proof span, at the lifting of F^n -> H^n(K) to F^n[-n] -> K; that phrase is kept verbatim and the surrounding argument is complete without it. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Perfect objects as bounded complexes with finite cohomology, and strong generation of a triangulated category. invoked at proof lines 20827-20831. Fixes the two notions the statement relates and the generation index that the induction tracks.
- Finite global dimension of a regular ring and subadditivity of projective dimension in a short exact sequence. invoked at proof lines 20836-20837 and 20875-20877. Supplies the bound d and is what makes the projective dimension of H^n(K') drop by one.
- Vanishing of higher Ext between projective modules and the splitting of a complex into the shifts of its cohomology. used at proof lines 20845-20851. Is the base case of the induction.

Source reference. Original lemma 065S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7911–7911. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00OE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 892–892. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 09SJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12357–12357. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GM4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20744–20744. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 19115: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19115–19115. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a Noetherian regular ring of finite dimension. Then R is a strong generator for the full subcategory $D_{perf}(R) \subset D(R)$ of perfect objects.*

Proof. We will use that an object K of $D(R)$ is perfect if and only if K is bounded and has finite cohomology modules, see Lemma 19115. Strong generators of triangulated categories are defined in Derived Categories, Definition 09SJ. Let $d = \dim(R)$.

Let $K \in D_{perf}(R)$. We will show $K \in \langle R \rangle_{d+1}$. By Algebra, Lemma 00OE every finite R-module has projective dimension $\leq d$. We will show by induction on $0 \leq i \leq d$ that if $H^n(K)$ has projective dimension $\leq i$ for all $n \in \mathbf{Z}$, then K is in $\langle R \rangle_{i+1}$.

Base case $i = 0$. In this case $H^n(K)$ is a finite R-module of projective dimension 0. In other words, each cohomology is a projective R-module. Thus $\mathrm{Ext}^i_R(H^n(K), H^m(K)) = 0$ for all $i > 0$ and $m, n \in \mathbf{Z}$. By Derived Categories, Lemma 0GM4 we find that K is isomorphic to the direct sum of the shifts of its cohomology modules. Since each cohomology module is a finite projective R-module, it is a direct summand of a direct sum of copies of R. Hence by definition we see that K is contained in $\langle R \rangle_1$.

Induction step. Assume the claim holds for $i < d$ and let $K \in D_{perf}(R)$ have the property that $H^n(K)$ has projective dimension $\leq i + 1$ for all $n \in \mathbf{Z}$. Choose $a \leq b$ such that $H^n(K)$ is zero for $n \notin [a, b]$. For each $n \in [a, b]$ choose a surjection $F^n \to H^n(K)$ where F^n is a finite free R-module. Since F^n is projective, we can lift $F^n \to H^n(K)$ to a map $F^n[-n] \to K$ in $D(R)$ (small detail omitted). Thus we obtain a morphism $\bigoplus_{a \leq n \leq b} F^n[-n] \to K$ which is surjective on cohomology modules. Choose a distinguished triangle

$$K' \to \bigoplus\nolimits_{a \leq n \leq b} F^n[-n] \to K \to K'[1]$$

in $D(R)$. Of course, the object K' is bounded and has finite cohomology modules. The long exact sequence of cohomology breaks into short exact sequences

$$0 \to H^n(K') \to F^n \to H^n(K) \to 0$$

by the choices we made. By Algebra, Lemma 065S we see that the projective dimension of $H^n(K')$ is $\leq \max(0, i)$. Thus $K' \in \langle R \rangle_{i+1}$. By definition this means that K is in $\langle R \rangle_{i+1+1}$ as desired. $\square$